Arpit Sharma

BTech CSE (AI & ML) — GLA University — Graduating 2028

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EDUCATION

Bachelor of Technology in Computer Science (AI & ML) GLA University, Mathura, UP
Expected May 2028 CGPA: 6.76/10

EXPERIENCE

Full Stack & AI/ML Project Developer Personal Projects — GitHub
June 2024 – Present

- Engineered **Voice-Enabled Accessibility Assistant** in Python using SpeechRecognition, OpenAI APIs, and Computer Vision for blind and visually impaired users; deployed production version on Render with real-time object detection and voice narration
- Implemented object identification system providing audio descriptions of surroundings; application assists users with navigation, environment awareness, and accessibility–enabling independent mobility
- Integrated multi-modal AI pipeline combining speech recognition, LLM inference, and text-to-speech synthesis; optimized response latency for real-time accessibility use cases
- Developed and published open-source voice assistant with comprehensive documentation on GitHub; actively maintaining codebase and iterating on user feedback

Machine Learning Engineer - Project Based Academic & Self-Directed Learning
Jan 2024 – Present

- Built **Breast Cancer Detection ML Model** using Python, scikit-learn, and data classification techniques; trained supervised learning classifier achieving robust performance on medical imaging datasets
- Implemented IoT project integrating Python backend, sensor data processing, and embedded systems communication for real-world hardware interaction
- Created interactive data visualization dashboard using Streamlit and Python; deployed college project with interactive UI for data exploration and model insights

PROJECTS

Voice-Control Assistant for Accessibility Python — SpeechRecognition — OpenAI API — Computer Vision — Render
Deployed & Ongoing

- Full-stack voice interface application enabling visually impaired users to navigate physical environments through audio feedback and real-time object identification
- Implemented multi-API integration: speech-to-text (SpeechRecognition), AI responses (OpenAI), and visual analysis (computer vision models) in unified Python architecture
- Deployed production-ready application on Render; handles multi-user requests with optimized latency for accessibility-critical use cases
- Demonstrates end-to-end AI/ML pipeline: input processing, model inference, and real-time response generation
- GitHub: github.com/PILOT-SHARMA/Voice-control-assistant

Breast Cancer Classification ML Model Python — scikit-learn — Data Science — Machine Learning
Completed

- Developed supervised learning classification model for medical imaging analysis; implemented data preprocessing, feature engineering, and model evaluation
- Explored multiple ML algorithms; evaluated performance metrics and model interpretability for healthcare applications
- Applied industry-standard ML workflow: data exploration, train-test splitting, hyperparameter tuning, and cross-validation

Voice Recognition & Speaker Identification System Python — Audio Processing — ML
Completed

- Engineered audio classification system distinguishing between different speakers using acoustic features and machine learning
- Implemented signal processing pipeline for audio feature extraction and voice identification

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL, JavaScript, HTML, CSS
AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, OpenAI API, Computer Vision
Full Stack: React, Node.js, Express.js, Streamlit, Django
Databases: SQL, MongoDB
Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Linux
Cloud & Deployment: Render, Azure (Certified), Docker basics
Other: Bootstrap, REST APIs, Data Visualization, Problem Solving

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft Azure AI Engineer Certified (AZ-101)** – Demonstrates expertise in Azure AI services, machine learning pipelines, and cloud-based AI solutions
- **Microsoft Azure Administrator Certified (AZ-900)** – Foundational cloud computing and Azure platform expertise
- **Participant – Technex IIT BHU Hackathon** – Competed in premier national hackathon; developed and pitched AI/tech solutions
- **4+ Years Programming Experience** – Started coding in class 11; progressed from Python basics through advanced ML/AI applications; consistent learner and builder
- **Active GitHub Contributor** – 19 public repositories showcasing diverse projects in AI, full stack development, and open-source participation

INTERESTS & GOALS

Focus Areas: AI Engineering, Full Stack Development, Machine Learning, Accessibility Technology, Cloud Computing
Career Goal: Secure internship in AI/ML or Full Stack Development at FAANG companies; contribute to impactful AI projects and scalable systems
Learning: Actively expanding expertise in Deep Learning, Generative AI, and advanced system design